Improving Reasoning and Generalization in Small LM through a Hybrid Reward Approach with GRPO

Author information scrubbed for double-blind reviewing

No Institute Given

Abstract. Improving the reasoning capabilities of small language models (LMs) through reinforcement learning requires balancing external factual accuracy with internal model calibration. External rewards based on verifiable correctness provide strong grounding, but are sparse and task-specific. Internal confidence-based rewards are dense and scalable, but prone to miscalibration and overconfident errors. Thus, relying exclusively on either signal may lead to instability or limited generalization, particularly in small LMs where limited capacity amplifies reward noise and training sensitivity. We propose GRPO-Hybrid, a dynamic reward scheduling framework within GRPO that integrates external ground-truth verification with internal self-certainty signals. Training begins in a correctness-dominated regime to enforce alignment with ground-truth objectives and progressively increases the weight of calibrated internal confidence to encourage coherent and efficient reasoning trajectories. The hybrid reward combines correctness, self-certainty, calibration, and process-level signals through a temporally adaptive scheme tailored to small LMs. Experiments with a 4B-parameter model on mathematical reasoning and out-of-domain benchmarks show that GRPO-Hybrid consistently outperforms single-reward baselines and static reward partitions. Beyond accuracy gains, we observe a strong correlation between correctness and computational efficiency: the hybrid model reaches correct solutions using significantly fewer tokens, indicating more direct reasoning paths. Additional results on multi-turn tool-use tasks further demonstrate that GRPO-Hybrid provides a stronger initialization for iterative reinforcement learning. These findings show that dynamically coupling external verification with internal calibration is an effective and principled strategy for enhancing robustness and efficiency in small LMs. Implementation and code: https://anonymous.4open.science/r/GRPO_Hybrid_reward-0D4D

Keywords: Reinforcement Learning · GRPO · LLM

1 Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities in generating coherent and contextually appropriate text across diverse domains [1,10]. However, aligning LLM outputs with desired reasoning behaviors remains challenging, particularly for tasks requiring compositional reasoning or handling out-of-distribution inputs. While Supervised Fine-Tuning (SFT) [9] instills task-specific knowledge, it often fails to guarantee alignment with complex objectives or generalization beyond the training distribution [19]. Reinforcement Learning from Human Feedback (RLHF) [8] has emerged as an effective paradigm for guiding LLMs toward desired behaviors by optimizing reward functions reflecting human preferences or task-specific criteria. Standard approaches such as Proximal Policy Optimization (PPO) [12] and Direct Preference Optimization (DPO) [11] enable learning from reward signals but suffer from instability, sensitivity to reward design, and high computational costs. Group-Relative Policy Optimization (GRPO) [4] addresses these limitations by normalizing reward signals across groups of model completions, providing improved stability and robustness. Instead of relying solely on a learned value function as a baseline, GRPO computes advantages by ranking or normalizing rewards within each group, thus reducing variance and mitigating reward-scale sensitivity. This group-wise normalization enables more stable updates and has proven particularly effective in reasoning-intensive tasks. A central question in recent literature concerns the source of the reward signal used within GRPO-based training pipelines. Two main paradigms have emerged. The first relies on an external reward model, where feedback is provided by a separately trained evaluator or by verifiable metrics. Thus, works such as DeepSeek-R1 [4] employ reward models based on answer ground truth and format correctness. In this setting, the reward signal is derived from objective criteria, such as whether the final response matches a verified solution and whether it adheres to a predefined output structure. The second approach is to use the model's internal confidence as a reward signal. This is the direction pursued by Intuitor [18], where self-certainty estimates are exploited to score answer quality. In this framework, the model evaluates its own outputs by assigning confidence-based signals that reflect the consistency and reliability of its reasoning process. This internal feedback mechanism removes the need for an external reward model and enables a more self-contained training loop. Although, it also introduces challenges related to calibration and the risk of reinforcing overconfident but incorrect predictions.

Our Contribution

To address the complementary limitations of external and internal reward signals, we introduce a hybrid dynamic reward schedule within the GRPO framework. External rewards, derived from verifiable correctness and structural constraints, provide a stable optimization signal but are typically sparse and narrowly defined. Internal confidence-based rewards offer dense, model-aware feedback that can capture reasoning consistency, yet they are susceptible to miscalibration and self-reinforcement. Exclusive reliance on either signal constrains learning: the former limits exploration beyond predefined objectives, while the latter risks amplifying systematic errors.

These challenges are particularly relevant in *small/medium-sized* large language models, where limited capacity increases sensitivity to reward noise and training instability. In such regimes, reward design plays a central role in shaping reasoning behavior, and improvements cannot be attributed to scale alone. Thus, propose a temporally adaptive weighting mechanism that reconciles objective supervision with model-intrinsic feedback. Training is initially guided by the external reward to ensure alignment with ground-truth objectives and stable policy updates. As optimization progresses, the weighting gradually shifts toward the internal reward, encouraging coherent reasoning trajectories and structured exploration. The formal specification of this schedule is detailed in Section 3.3. This dynamic coupling departs from static single-source reward designs and provides a principled approach to reward composition under constrained model capacity.

We evaluate the proposed hybrid GRPO approach on Qwen3-4B, initialized with supervised fine-tuning and optimized using the hybrid schedule. Experiments span mathematical reasoning under varying dataset scales, out-of-domain generalization, and multi-tool API reasoning requiring compositional execution. Across settings, the hybrid formulation consistently outperforms single-reward baselines, demonstrating that carefully balanced reward signals enable robust reasoning improvements even in moderate-scale models.

The remainder of this paper is organized as follows. First, Section 2 reviews the current state-of-the-art work on GRPO techniques and analyzes existing challenges. Section 3 then details the methodology behind our proposed Hybrid approach, followed by Section 4, which describes the experimental setup to ensure reproducibility. Next, Section 5 presents an analysis of the results obtained by testing multiple model versions. Finally, Section 6 discusses these findings and proposes possible directions for future development or extensions of this work.

2 Related Work

Group Relative Policy Optimization The DeepSeek-R1 technical report [4] introduced Group Relative Policy Optimization as a scalable and stable alternative to traditional RL algorithms for training reasoning models. GRPO addresses key limitations of PPO by eliminating the need for a separate value network, instead normalizing rewards within groups of sampled outputs to reduce variance and improve training stability. This group-based normalization enables the algorithm to extract more informative learning signals even from sparse or noisy rewards, while significantly reducing computational overhead compared to actor-critic methods. DeepSeek-R1 demonstrated that GRPO, combined with verifiable rewards for mathematical and coding tasks, achieves competitive performance with leading reasoning models while requiring substantially fewer computational resources. The method's efficiency stems from its ability to perform policy updates using only sampled trajectories and their relative quality within each group, making it particularly suitable for training large-scale models on reasoning tasks. However, GRPO's effectiveness is closely tied to the quality and design of the reward function, and its group-based comparison mechanism may be sensitive to the diversity and size of sampled groups, potentially affecting convergence in scenarios with highly imbalanced or homogeneous outputs.

Reinforcement Learning from Human Feedback (RLHF) Reinforcement Learning from Human Feedback (RLHF) has emerged as a foundational technique for refining LLMs to align with human values and preferences. The seminal work [8] demonstrated that training reward models on human preference data could effectively guide model behavior. However, the resource demands of collecting extensive human annotations present significant scalability challenges [15]. To address these limitations, methods such as Direct Preference Optimization (DPO) [11] were developed to streamline the alignment process by eliminating the need for explicit reward model training. Despite these improvements, approaches relying on human or model-approximated preferences remain susceptible to biases inherent in the reward modeling process.

Reinforcement Learning with Verifiable Rewards (RLVR) An alternative paradigm, Reinforcement Learning with Verifiable Rewards (RLVR), has gained prominence in domains where objective correctness criteria exist, such as mathematical problem-solving and program synthesis [4,17,6]]. By leveraging deterministic verification mechanisms—including exact answer comparison and unit test execution [4,7] RLVR circumvents the challenges associated with learned reward models. Recent breakthroughs, exemplified by systems like DeepSeek-R1 [4], have demonstrated exceptional reasoning performance through RLVR combined with advanced optimization techniques such as GRPO and its extensions. However, RLVR's effectiveness remains constrained to tasks with well-defined verification procedures, and its emphasis on outcome-based signals may hinder transfer to domains requiring process-level reasoning or lacking clear ground-truth solutions.

Learning to Reason without External Rewards Recent work has explored the possibility of developing reasoning capabilities without relying on external verification or human feedback. The Intuitor framework [18] introduces a self-supervised approach where models learn to reason by generating their own training signals through iterative self-improvement. Rather than depending on ground-truth answers or human preferences, Intuitor leverages the model's internal consistency and coherence as implicit reward signals, enabling the discovery of reasoning patterns through exploration and self-evaluation. This paradigm shift demonstrates that LLMs can develop sophisticated reasoning abilities by learning from their own generated reasoning traces, assessed through internal verification mechanisms. While this approach offers promising advantages in terms of scalability and independence from external supervision, it faces challenges in ensuring alignment with correct solutions and preventing the reinforcement of flawed reasoning patterns. The method's reliance on the model's intrinsic evaluation capabilities raises questions about generalization to domains where internal consistency may not correlate with correctness, and about the potential for divergence from human-aligned reasoning strategies without external grounding signals.

3 Method

In this section, we present the **Hybrid Reward Approach**, a novel reinforcement learning framework that integrates both external reward signals—derived from verifiable correctness criteria—and intrinsic reward signals based on the model's self-certainty, enabling improved reasoning capabilities and enhanced out-of-domain generalization in small LMs. We start by singularly describing the two existing rewards components (external and internal) and we analized their limitations, which motivates this work. We then describe how we design the Hybrid approach (3.3) and in particular how we mixed the two reward signals.

3.1 From External feedback

Modern strategies for refining Large Language Models (LLMs) through reinforcement learning generally bifurcate into two distinct methodologies: Reinforcement Learning from Human Feedback (RLHF) and Reinforcement Learning from Verifiable Rewards (RLVR). In the RLHF framework [8,20], the primary goal is to synchronize the model policy π_θ with nuanced human expectations, which are typically modeled by a proxy reward function r_ϕ. The mathematical objective is defined as:

$$\max_{\pi_\theta} \mathbb{E}_{o\sim\pi_\theta(q)}\left[r_\phi(q,o) - \beta D_{KL}(\pi_\theta(o|q) \parallel \pi_{ref}(o|q))\right] \tag{1}$$

where q is an input query, o is the generated output, π_{ref} is an initial reference policy, and β is a coefficient controlling the KL divergence to prevent excessive deviation from π_{ref} Conversely, RLVR bypasses the subjective ambiguity of human preference by employing deterministic and objective verifiers, particularly in domains like mathematics or programming where correctness is non-negotiable. In RLVR, the reward R_{ver} is binary or step-wise verifiable (e.g., unit test passes or final answer matching), effectively eliminating the "reward hacking" [13] often observed in RLHF's proxy models. In this context, reward hacking refers to the tendency of models trained with proxy or imperfect reward signals to exploit loopholes in the reward function, maximizing the measured score without genuinely improving task performance. By utilizing a verifiable signal $V(y)$, the objective shifts toward a grounded reinforcement signal:

$$\max_{\pi_\theta} \mathbb{E}_{o\sim\pi_\theta(q)}\left[v(q,o) - \beta D_{KL}(\pi_\theta(o|q) \parallel \pi_{ref}(o|q))\right] \tag{2}$$

where v(q, o) is a verifiable reward function. Unlike the scalar output of a neural reward model, $v(q,o)$ provides a grounded signal based on logical correctness (e.g., compiler success or ground-truth matching). In this framework, the reward is sparse but high-fidelity, ensuring that the model converges toward computationally valid solutions rather than merely linguistically plausible ones. This distinction is crucial for scaling reasoning, because the truth-value of an output can be programmatically confirmed.

3.2 From Internal Feedback (RLIF)

The primary bottleneck in scaling Large Language Models (LLMs) via reinforcement learning is the dependency on external signals, whether in the form of human preference labels (RLHF) or deterministic verifiers (RLVR). Reinforcement Learning from Internal Feedback (RLIF) emerges as a self-contained alternative where the model leverages its own latent knowledge to drive policy improvement. Under the RLIF paradigm, the optimization objective becomes:

$$\max_{\pi_\theta} \mathbb{E}_{o \sim \pi_\theta(q)} \left[v(q, o) - \beta D_{KL}(\pi_\theta(o|q) \parallel \pi_{ref}(o|q)) \right] \tag{3}$$

where u(q, o) represents an intrinsic signal derived from the model's internal state or computation, rather than external verification. The key challenge lies in identifying intrinsic signals that correlate with output quality and can effectively guide learning. In the RLIF work[18], they propose INTUITOR, a novel RLIF method that utilizes a model's own confidence as the sole intrinsic reward signal u(q, o). They adopt the self-certainty metric[5], defined as the average KL divergence between a uniform distribution U over the vocabulary V and the model's next-token distribution:

$$\textbf{Self-certainty}(o|q) := \frac{1}{|o|} \sum_{i=1}^{|o|} \mathrm{KL}(U \| p_{\pi_\theta}(\cdot|q, o_{<i})) \tag{4}$$

This is equal to:

$$\textbf{Self-certainty}(o|q) := -\frac{1}{|o| \cdot |V|} \sum_{i=1}^{|o|} \sum_{j=1}^{|V|} \log(|V| \cdot p_{\pi_\theta}(j|q, o_{<i})) \tag{5}$$

where $o < i$ are the previously generated tokens and $p(j|q, o < i)$ is the model's predicted probability for token j at step i. Higher self-certainty values indicate greater confidence. The most significant risk identified in the Intuitor paper is the phenomenon of overconfident hallucinations. Since the reward is derived from the model's own confidence scores, a model that generates an incorrect answer with high statistical probability (often due to biases inherited from the pre-training data) will be positively reinforced for its error. This creates a feedback loop where the model consolidates "logical dead-ends", effectively becoming more certain about incorrect reasoning paths. This can happen especially in difficult tasks such as mathematical problem solving (which is the domain of our training dataset). According to the creators of the Intuitor paper, this phenomenon can occur in up to 5% of the cases, leading to the so-called Overconfidence Failure problem, caused by the absence of External Grounding. Unlike classic external reward signal approaches, which utilize exact-match checks and numerical validators, Intuitor lacks an external "anchor" to reality. Intuitor's reliance on internal consistency as a proxy for truth can lead to Reward Hacking, where the model learns to optimize for high-confidence linguistic patterns that satisfy its internal reward but fails to produce a valid execution payload.

Table 1: Evolution of Reward Component Weights

Component	**Function** $\omega(t)$	**Initial** $(t=0)$	**Final** $(t=T)$
Answer Correctness(w_{corr})	$0.40 - 0.20\phi(t)$	0.40	0.20
Self-Certainty(w_{cert})	$0.20 + 0.20\phi(t)$	0.20	0.40
Calibration(w_{cal})	0.25	0.25	0.25
Process Quality(w_{proc})	0.15	0.15	0.15

3.3 Hybrid Reward Approach

We now present **Hybrid Reward Approach**, a novel reinforcement learning framework that integrates both external reward signals—derived from verifiable correctness criteria—and intrinsic reward signals based on the model's self-certainty, enabling improved reasoning capabilities and enhanced out-of-domain generalization. The intuition was to adopt a dynamic reward scheme(1), in which the relative importance of the two rewards changes over time: as the training progresses, the contribution of the internal reward to the total reward is gradually increased, while the importance of the external reward decreases correspondingly. In other words, the training starts in a "grounded" regime dominated by external feedback and smoothly transitions to a more "introspective" regime, where internal reward becomes the main driver. This will help the model first learn correct behaviors and factual alignment, and later refine its reasoning processes and explore different reasoning paths without losing external coherence. At the core of the reward system, the **Self-Certainty** component measures the model's internal confidence in its generated response. Initially assigned a relatively low weight (ranging from 20% to 40%), the influence of Self-Certainty is gradually increased over the course of training. This progressive emphasis is governed by a normalized factor $\phi(t)$:

$$\phi(t) = \frac{t}{T_{max}}, \quad \phi(t) \in [0,1] \tag{6}$$

where t denotes the current epoch and T_{max} represents the total number of epochs scheduled for the training process. This factor acts as a non-stationary scheduler for the reward components. The factor $\phi(t)$ in the late training stages encourages the model to rely more on its internal reasoning signals rather than external supervision, thereby promoting improved out-of-distribution generalization and reducing dependence on task-specific evaluators.

As training progresses, the weight assigned to **Answer Correctness(**w_{corr}**)** is gradually reduced to approximately 20%, mitigating the risk of overfitting to the training distribution and preventing excessive reliance on explicit supervision. To further refine the interaction between confidence and correctness, a **Calibration** reward component is introduced with a fixed weight of 25%. This component (ranging from 0 to 1) assigns a score equal to the Self-Certainty when the answer is correct; otherwise, it returns 1.0 - Self-Certainty. The final component is **Process Quality**, a reward term that evaluates whether the generated response contains logical connectors such as “therefore,” “because,” “since,” “thus,” and similar expressions, which indicate structured reasoning.

As previously said we used the GRPO training technique. The core idea behind the GRPO optimization is to sample multiple candidate outputs for a given query and use their relative rewards to estimate advantages for policy updates. For each query $q \sim P(Q)$, GRPO samples a group of G outputs $o_1, ..., o_G$ using a behavior policy $\pi_{\theta_{old}}$ (it can be a previous iteration or the base SFT model). The target policy π_θ is then optimized by maximizing:

$$\begin{aligned}\mathcal{J}_{GRPO}(\theta) = \mathbb{E}[q \sim P(Q), \{o_i\}_{i=1}^{G} \sim \pi_{\theta_{old}}(O|q)]\\ \frac{1}{G}\sum_{i=1}^{G}\left(\min\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}A_i,\ \text{clip}\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}, 1-\varepsilon, 1+\varepsilon\right)A_i\right)\right.\\ \left.-\beta\,\mathbb{D}_{KL}\left(\pi_\theta \| \pi_{ref}\right)\right), \end{aligned} \tag{7}$$

$$\mathbb{D}_{KL}\left(\pi_\theta \| \pi_{ref}\right) = \frac{\pi_{ref}(o_i|q)}{\pi_\theta(o_i|q)} - \log\frac{\pi_{ref}(o_i|q)}{\pi_\theta(o_i|q)} - 1, \tag{8}$$

Hyperparameters ϵ (for clipping) and β (for KL penalty strength) control stability and exploration, and A_i is the advantage estimate.

Integration of Hybrid Reward The key innovation in this work is the use of a hybrid reward, formed by combining internal and external signals, within GRPO's advantage computation. Specifically, each output o_i is scored by:

$$r_i = w_{\text{cert}} \cdot s_{\text{cert}} + w_{\text{corr}} \cdot s_{\text{corr}} + w_{\text{cal}} \cdot s_{\text{cal}} + w_{\text{proc}} \cdot s_{\text{proc}} \tag{9}$$

$$\hat{A}_i = \frac{r_i - \mu_r}{\sigma_r + \epsilon}, \quad \mu_r = \frac{1}{G}\sum_{i=1}^{G} r_i, \quad \sigma_r = \sqrt{\frac{1}{G}\sum_{i=1}^{G}(r_i - \mu_r)^2} \tag{10}$$

4 Experimental Setup

Training Setup We utilized the Qwen3-4B model as our foundational backbone. This selection is motivated by two primary factors: first, its release as an open-weight checkpoint ensures full reproducibility and facilitates a granular analysis of training dynamics. Second, the Qwen3 architecture has demonstrated significant algorithmic efficiency during Reinforcement Learning (RL) stages. Specifically, the model family exhibits high policy stability and a well-calibrated initial logit distribution, which are critical for the convergence of Group Relative Policy Optimization (GRPO) in complex reasoning domains.
Starting from Qwen3-4B-Base we perform a preliminary Supervised Fine-Tuning training step with 5000 examples taken form the training split of the unsloth OpenMathReasoning-mini dataset, a high-density, curated collection of mathematical problems designed specifically for training reasoning-capable Large

Language Models. It serves as a distilled version of larger math corpora, optimized for efficiency and logical clarity. We use this Qwen3-4B-SFT model as backbone model.
All the GRPO-trained models presented in the sections below are trained with the same framework with 1000 examples taken form the training split of the open-r1/DAPO-Math-17k-Processed dataset.This dataset comprises $17,000$ high-quality mathematical samples curated for the Open-R1 project. Each sample $s_i = (q_i, a_i)$ consists of a natural language mathematical query q_i and a structured ground-truth solution a_i. This dataset serves as the foundational environment for our GRPO-based hybrid reward training, providing the necessary complexity to evaluate both numerical accuracy and logical reasoning consistency. Each update processes 16 problems, generating 6 candidate solutions per problem, with a default KL penalty of $\beta = 0.005$. All GRPO experiments are conducted using a fixed training configuration to ensure comparability across different experimental conditions. Key hyper-parameters include the learning rate, batch size, number of training steps, and the number of completions sampled per prompt (K). Sampling parameters such as temperature and top-p are kept constant throughout training. Different experimental runs vary only in the factors under investigation, namely the design of the reward function.

Evaluation Evaluations use the same chat-style prompting format as in training. All reported results are obtained using the same codebase and evaluation pipeline, ensuring consistency across experimental conditions. Additional implementation details and hyperparameter settings are provided in the following sections. For hardware usage, all training and experiments are run on a Nvidia A100 SXM4 80GB GPU. Model performance is evaluated using task-appropriate metrics that reflect both correctness and reasoning quality. For mathematical reasoning tasks,accuracy-based metrics are employed, including exact match accuracy. The performance of each model is also evaluated based on the quality of its responses using an LLM-Judge approach. In this step, a more sophisticated large language model with a greater number of parameters (Qwen3-80B) and specific reasoning-quality fine-tuning is queried to assess the responses. The choice of this specific model was driven by hardware constraints, as it represented the largest architecture compatible with our available computational resources. The model assigns a score on a 10-point scale reflecting overall quality, considering conciseness, clarity of reasoning, and completeness. This metric will be called Judge Score for the rest of the work. In addition, we define an efficiency metric to evaluate the trade-off between performance and verbosity, computed as **efficiency** = (average judge score $\times$ 100)/average number of tokens

We want to test the performance of the model in the out-domain field so we decided to use 3 different no-mathematical dataset: google/Boolq[2], tau/Commonsense_qa[14] and ARC-Challenge[3]. **BoolQ** serves as a rigorous test of reading comprehension through binary yes/no questions derived from naturally occurring search queries; it requires the model to perform nuanced inference based on specific Wikipedia passages. In contrast, **Commonsense_qa** shifts the focus toward everyday logic and general world knowledge by providing

Table 2: Performance comparison of various models on the out-domain benchmarks.

Models	Accuracy	Judge Score	Efficiency	Dataset
GRPO-Classic	84%	8.03	1.10	Boolq
	57%	7.73	0.82	Commonsense_qa
	63%	8.56	0.91	ARC-Challenge
GRPO-Intuitor	**87%***	**8.14***	**1.13***	Boolq
	56%	**7.93***	0.83	Commonsense_qa
	66%	8.43	0.89	ARC-Challenge
GRPO-DualHead	83%	8.07	1.10	Boolq
	57%	7.89	0.81	Commonsense_qa
	68%	8.41	0.90	ARC-Challenge
GRPO-Hybrid	83%	8.12	**1.13***	Boolq
	62%*	7.90	**0.84***	Commonsense_qa
	69%*	**8.59***	**0.91***	ARC-Challenge

multiple-choice questions where the correct answer must be distinguished from closely related semantic distractors. Finally, the **ARC-Challenge** provides a sophisticated assessment of scientific literacy, utilizing a subset of grade-school science questions that are specifically designed to be unsolvable by simple keyword matching, thus forcing the model to engage in complex, multi-step reasoning. Together, these datasets provide a comprehensive overview of how a model handles contextual inference, intuitive logic, and formal scientific knowledge in out-domain scenarios.

5 Results and Analysis

Starting from the intuition of balancing internal and external reward signals, we investigate the optimal design to enhance both the reasoning quality and efficiency of the model, specifically focusing on its generalization performance in out-of-domain testing. In this section, we initially discuss the results of a fixed partition schema(5.1) between the reward components, subsequently transitioning to the final design which employs a dynamic reward schema(5.2)

5.1 Hybrid Fixed Partition(GRPO-DualHead)

Based on the intuition of combining internal (Self-Certainty) and external (Answer Correctness) reward signals, we initially designed a fixed partition between the reward components. We evaluated three different static reward schemes:

- **GRPO-DualHead(30/70)** in which we allocated 30% of the weight to internal rewards and 70% to external ones
- **GRPO-DualHead(50/50)** in which equal weight was assigned to both reward signals

Table 3: Performance comparison of GRPO-DualHead models on the out-domain benchmarks.

Models	**Accuracy**	**Judge Score**	**Efficiency**	**Dataset**
DualHead(30/70)	**83%***	7.96	1.08	Boolq
	59%*	7.83	**0.83***	Commonsense_qa
	69%*	**8.52***	0.88	ARC-Challenge
DualHead(50/50)	**83%***	8.01	**1.12***	Boolq
	57%	7.81	0.80	Commonsense_qa
	69%*	8.49	0.86	ARC-Challenge
DualHead(70/30)	**83%***	**8.07***	1.10	Boolq
	57%	**7.89***	0.81	Commonsense_qa
	68%	8.41	**0.90***	ARC-Challenge

– **GRPO-DualHead(70/30)** in which we allocated 70% of the weight to internal rewards and 30% to external ones

From Table 3 we can observe that:

– In terms of accuracy, the results are relatively consistent across the board. However, as anticipated, the model that prioritizes external reward (DualHead(30/70))—specifically based on ground-truth correctness—achieves the highest performance across all test datasets. This indicates a direct correlation between the weight assigned to the external signal and the model's overall accuracy.
– Regarding answer quality—specifically completeness, reasoning depth, and clarity, as measured by the Judge Score—our results show a diverging trend. In the BoolQ and CommonsenseQA benchmarks, performance improved as the weight of the internal reward signal increased. In contrast, we observed the opposite behavior with the ARC-Challenge dataset, where a higher reliance on external signals yielded better scores. This behavior can be explained by the nature of the datasets themselves. BoolQ and Commonsense-QA focus primarily on linguistic inference and semantic logic rather than hard scientific facts. Therefore, a model trained predominantly to develop a qualitative and consistent reasoning path outperforms a model focused on external (ground-truth-based) validation. Conversely, in the ARC-Challenge—a dataset composed of grade-school science questions where specific knowledge is often required—a model over-reliant on internal logic may yield poorer results if it lacks factual accuracy.
– Regarding efficiency, we observed similar results across all models and test datasets. There is no evidence to suggest that the varying levels of importance assigned to the two reward signals lead to different behaviors in terms of computational performance.

These results (Table 3) highlight the potential of combining the two reward components. However, we also recognize that a fixed partitioning schema has

Table 4: Comparison of efficiency performance across multiple benchmarks. The Hybrid approach consistently achieves superior results in reasoning-intensive tasks.

Model	Method	BoolQ	CSQA	ARC-C
Qwen3-4B	SFT	0.58	0.46	0.52
	GRPO-Classic	1.10	0.82	0.91
	GRPO-Hybrid	**1.13***	**0.84***	**0.91***

inherent limitations. To bridge the gap—creating a model with both a robust knowledge base and the ability to produce qualitatively consistent and logical answers—we developed Hybrid Dynamic Scheduling

5.2 Hybrid Dynamic Scheduling (GRPO-Hybrid)

From the results (Table 2), we can observe that the Hybrid model achieves the best performance in terms of efficiency, attains the highest accuracy on two out of the three datasets, obtains the best judge score on the third dataset, and shows results comparable to the best-performing method on the remaining two datasets. Notably, in terms of both computational and time resources, it achieves the best overall efficiency across all datasets. For what concern efficiency, we can also observe, as illustrated in Table 4, that the GRPO technique—and specifically the GRPO-Hybrid approach—dramatically improves model efficiency compared to the base SFT model. Based on the results presented (Table 2), we can draw the following observations:

Solving the Absence of External Grounding problem Unlike GRPO-Classic approach, which utilizes exact-match checks and numerical validators, Intuitor lacks of an external "anchor" to reality. Intuitor's reliance on internal consistency as a proxy for truth can lead to Reward Hacking [13], where the model learns to optimize for high-confidence linguistic patterns that satisfy its internal reward but fail to produce a valid execution payload. Since the reward is derived from the model's own confidence scores, a model that generates an incorrect answer with high statistical probability (often due to biases inherited from the pre-training data) will be positively reinforced for its error. The GRPO-Hybrid model addresses this challenge by integrating a dual-reward mechanism: an external ground-truth signal derived from automated answer verification, and the calibration-based reward. This secondary component reinforces the policy to produce correct answers while simultaneously maximizing model confidence. Empirical results demonstrate that this approach improved accuracy across two out of three benchmarks while either enhancing or maintaining the quality of reasoning, as evidenced by the Judge Score results.

Efficiency as a Proxy for Logical Coherence The GRPO-Hybrid model achieves superior efficiency scores across all three benchmarks (see Table 4), indicating a more optimized reasoning trajectory. We consider this a primary metric, as a high efficiency score demonstrates that the model does not rely on exhaustive or repetitive trial-and-error to reach a correct solution. Instead, it suggests the development of a refined, direct reasoning path. Our data reveals a strong correlation between accuracy and efficiency; specifically, the Hybrid model reaches correct solutions using significantly fewer tokens than the baseline. This suggests that advanced reasoning is characterized not by increased computation ('thinking more'), but by qualitative optimization ('thinking better'), where the model identifies the most direct logical path to reduce both latency and token consumption.

The power of the Dynamic reward scheme As illustrated in Table 2, the GRPO-Hybrid model consistently outperforms GRPO-DualHead across all evaluated benchmarks. As previously explained in Section 5.1, a fixed reward partitioning scheme may constrain a model's exploration of diverse reasoning paths. Conversely, it can lead to a lack of ground-truth validation, resulting in diminished performance in terms of accuracy. In contrast, the Dynamic scheme initially prioritizes the acquisition of robust, ground-truth-based knowledge. Once this foundation is established, the model is gradually encouraged to explore alternative reasoning trajectories, ultimately enhancing its overall logical and reasoning capabilities.

5.3 Ablation

To determine the initial optimal weights schema for our multi-component reward function in GRPO-Hybrid training, we developed a systematic weight optimization pipeline that operates through evaluation only, without performing any model training. Our reward function comprises four components—certainty (self-certainty), answer correctness, calibration, and process quality—each requiring careful weighting to balance competing objectives. We employed a two-stage optimization strategy on 1,000 examples from the DAPO-Math-17k-Processed dataset, split into 75% training, 15% validation and 10% out-of-distribution test sets. The first stage utilized Bayesian optimization with 50 trials to efficiently explore the weight space by building a Gaussian Process surrogate model, while the second stage performed multi-objective Pareto optimization with 50 trials to identify trade-off frontiers between accuracy, calibration, OOD generalization, and process quality. Starting from an initial configuration of self-certainty = 0.25, answer correctness = 0.40, calibration = 0.20, process = 0.15, the pipeline evaluated our Qwen-4B SFT model across different weight configurations using structured response formats with explicit reasoning and solution markers. This evaluation-only paradigm preserved the SFT model's learned capabilities while significantly reducing computational cost compared to full training runs. The pipeline generated comprehensive outputs including optimized weights in JSON format, ablation visualizations showing individual component sensitivity, Bayesian convergence plots, and Pareto frontier analysis for trade-off examination. These empirically validated weights were then directly integrated into our GRPO training script, ensuring that the multi-objective optimization results—balancing accuracy, reliability, generalization, and reasoning quality—informed the subsequent policy optimization phase with quantitative evidence rather than manual intuition.

Table 5: BFCL train results

Model	Average Reward	Perfect Match Rate
GRPO-Hybrid-BFCL	**0.54***	**50%***
GRPO-Classic-BFCL	0.47	42.6%

5.4 Hybrid scheme model as a training base

To further evaluate the capabilities of the proposed Hybrid approach, we use it as the base model for an additional GRPO training phase, in which we adopt the best hyperparameters identified in the previous section. The goal is to demonstrate that the Hybrid model is not only superior in out-of-domain evaluations, but also represents a better initialization point for subsequent training phases. The training dataset consists of the BFCL v3 multi-turn split. This task is particularly challenging as it involves multi-step API orchestration, requiring the model to maintain state and logical consistency across several turns. Using this benchmark allowed us to test the limits of the GRPO-Hybrid model in complex, real-world tool-use scenarios. We conducted an additional GRPO training stage, utilizing both the previously defined GRPO-classic and GRPO-Hybrid models as initial checkpoints. This iterative refinement was designed to further stabilize the policy and evaluate the cumulative performance gains across both architectures. Data from Table 5 indicates that the GRPO-Hybrid model provides a more robust initialization for iterative optimization. The significant variance in final performance suggests that the choice of the base model is a critical factor in the GRPO pipeline, where the Hybrid configuration consistently yields more favorable training dynamics

6 Discussion and Future Reasearch

Scalability and Generalization. Our experiments, constrained by computational resources, utilize relatively compact models trained on relatively small, unsupervised corpora. We aim to demonstrate the potential of mixing internal model confidence and external reward signal for policy optimization. The results demonstrate that this signal combination consistently promotes more coherent, logically justified, and efficient responses, even when applied to highly complex tasks. Future work could explore these benefits in larger or smaller foundation models and on more diverse, real-world datasets.

Alternative Internal Metrics. Future work should explore alternative internal confidence metrics beyond token-level certainty. While token probabilities provide a tractable measure of model confidence, they capture only surface-level uncertainty and may not reflect deeper epistemic doubt in the reasoning process. Richer intrinsic signals such as attention entropy, hidden state consistency, or layer-wise uncertainty measures could provide more faithful confidence estimates.

Investigating these alternatives may improve the alignment between model self-assessment and actual performance, leading to better-calibrated reasoning models.

Integration with Alternative Policy Gradient Frameworks While our current implementation utilizes GRPO to optimize self-certainty, the Hybrid approach is inherently adaptable to a broad range of policy gradient algorithms. The selection of GRPO was a strategic decision, aimed at balancing high performance with the intensive computational requirements of large-scale LLM fine-tuning. Crucially, the core principle—integrating self-certainty with ground-truth verification—is optimizer-independent. Its potential efficacy when paired with other frameworks, such as PPO[12] or REINFORCE[16], remains a promising direction for future empirical investigation.

7 Conclusion

In this paper, we introduced the GRPO-Hybrid model, a novel framework that integrates internal self-certainty signals with external ground-truth verification to optimize the reasoning capabilities of Small Language Models. Our experiments, conducted on the out-domain benchmark dataset and on the challenging BFCL v3 multi-turn benchmark, demonstrate that a dynamic reward partitioning scheme significantly outperforms the single reward signals reward scheme. Our findings yield three primary contributions to the field of Reinforcement Learning for LLMs. First, the GRPO-Hybrid approach demonstrates **Superior Reasoning Efficiency**; it does not merely improve accuracy but promotes a more optimized reasoning trajectory. By aligning high-confidence predictions with factual correctness, the model identifies more direct logical paths, resulting in a significant reduction in token consumption and computational latency. This leads to a **Dynamic Reward Synergy**, where we demonstrated that prioritizing ground-truth alignment in the early stages of training—followed by a gradual increase in exploratory freedom—is essential for developing robust reasoning skills without falling into the trap of reward hacking or repetitive trial-and-error. Finally, our results highlight the **Scalability and Adaptability** of this framework, showing that the Hybrid configuration serves as an exceptional foundation for iterative training pipelines. Furthermore, while we successfully employed GRPO for its computational efficiency, the Hybrid framework remains fundamentally algorithm-agnostic, offering a versatile blueprint for integration with other policy gradient methods such as PPO or REINFORCE.

Ultimately, this work proves that advanced AI reasoning is not a matter of "thinking more", but rather "thinking better". By fostering a synergy between internal calibration and external verification, we move closer to agents that are not only accurate but also computationally efficient and logically coherent. Future research will focus on expanding this hybrid signal to even more diverse multi-agent and multi-tool environments.

AI Usage Disclosure The authors used generative AI tools exclusively for editorial support (grammar checking and rephrasing). All scientific content, methodology, and conclusions were developed solely by the authors, who take full responsibility for the accuracy of the work.

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